

DUNHUANG, China

WMO: 524180

Lat: 40.150N

Lon: 94.683E

Elev: 3740

StdP: 12.82

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.1	3.5	-14.0	2.9	16.0	-10.5	3.6	15.9	17.6	20.7	15.7	20.3	1.9	180	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	95.9	64.8	93.5	63.8	91.2	62.7	68.6	89.2	66.9	87.1	65.4	85.2	5.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.4	96.8	75.1	60.4	90.1	75.1	58.4	84.0	75.4	35.3	89.6	33.7	87.0	32.5	85.5	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.3	12.1	DB	-5.4	100.9	3.8	2.5	-8.1	102.7	-10.3	104.2	-12.4	105.6	-15.2	107.4	(4)
(5)			WB	-6.1	73.0	3.7	2.0	-8.8	74.4	-10.9	75.6	-13.0	76.7	-15.6	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	18.5	29.2	43.3	56.9	67.0	75.2	78.6	75.6	65.4	51.0	36.2	21.8	(6)	
	DBStd	21.63	6.16	7.59	8.03	7.81	6.39	4.73	4.43	4.55	5.57	6.28	7.23	6.72	(7)	
	HDD50	3176	976	584	233	27	0	0	0	0	1	66	416	873	(8)	
	HDD65	6133	1441	1004	672	260	54	2	0	1	62	434	865	1338	(9)	
	CDD50	3784	0	0	26	234	528	755	887	795	462	97	0	0	(10)	
	CDD65	1265	0	0	0	17	116	306	422	331	73	0	0	0	(11)	
	CDH74	15119	0	0	4	311	1584	3595	4895	3710	972	48	0	0	(12)	
	CDH80	6857	0	0	0	73	567	1681	2504	1758	271	3	0	0	(13)	
(14) Wind	WSAvg	4.5	4.2	4.7	5.3	5.8	5.2	4.6	4.4	4.0	3.8	3.6	4.0	4.2	(14)	
(15) Precipitation	PrecAvg	1.5	0.0	0.0	0.0	0.1	0.1	0.3	0.5	0.2	0.1	0.0	0.0	0.0	(15)	
	PrecMax	4.1	0.2	0.2	0.4	0.6	0.5	1.4	2.4	1.1	0.5	0.2	0.5	0.1	(16)	
	PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	0.8	0.0	0.0	0.1	0.2	0.2	0.4	0.5	0.3	0.1	0.0	0.1	0.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.7	55.5	73.1	87.2	92.6	96.7	100.4	97.9	90.3	79.3	62.8	43.9	(19)	
		MCWB	30.1	37.8	46.9	54.2	57.3	62.5	66.6	66.5	60.4	52.5	43.4	32.1	(20)	
	2%	DB	36.6	50.1	67.9	81.3	89.3	93.9	96.9	94.4	86.1	74.3	57.9	39.5	(21)	
		MCWB	27.1	34.8	44.4	51.4	56.8	62.0	65.9	64.6	58.4	50.9	40.7	29.5	(22)	
	5%	DB	33.0	46.6	63.6	77.3	86.2	91.6	94.3	91.7	83.1	70.4	54.1	36.3	(23)	
		MCWB	24.9	33.1	42.4	49.6	55.3	61.1	65.5	63.8	57.4	48.9	39.0	27.7	(24)	
	10%	DB	29.9	43.0	59.6	73.5	83.0	89.2	91.7	89.2	79.9	66.5	49.9	33.2	(25)	
		MCWB	22.9	31.1	40.6	48.0	54.1	60.6	64.5	63.2	56.4	46.9	37.0	25.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.4	38.3	47.6	55.4	61.4	68.7	72.3	70.2	63.8	54.5	43.8	32.5	(27)	
		MCDB	40.8	53.9	71.6	83.7	86.5	88.4	93.4	91.0	81.8	73.1	61.1	42.7	(28)	
	2%	WB	27.5	35.4	44.9	52.3	58.2	65.9	69.4	67.8	61.2	51.9	41.1	29.8	(29)	
		MCDB	36.0	49.1	66.4	78.9	83.8	86.1	90.5	87.5	79.9	72.0	56.7	38.7	(30)	
	5%	WB	25.2	33.3	42.9	50.3	56.4	64.2	67.6	66.2	59.3	50.0	39.3	27.9	(31)	
		MCDB	32.5	45.8	62.8	75.0	81.9	84.9	88.4	85.4	77.6	68.6	53.6	35.7	(32)	
	10%	WB	23.2	31.2	40.8	48.5	54.9	62.7	66.2	64.8	57.7	47.9	37.2	25.9	(33)	
		MCDB	29.6	42.5	59.0	71.4	79.8	82.5	86.7	84.0	76.1	64.7	49.7	32.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	21.8	24.6	26.2	26.7	26.1	24.8	23.5	25.6	28.5	30.0	23.9	20.4	(35)	
		MCDBR	25.0	28.2	32.0	31.8	31.7	29.5	28.5	30.3	32.5	34.4	29.5	23.3	(36)	
	5% WB	MCWBR	17.7	17.3	15.9	13.0	12.0	11.4	11.7	12.0	12.3	14.8	16.0	15.5	(37)	
		MCWBR	24.2	27.2	30.9	29.3	27.7	25.2	24.9	26.7	27.7	32.7	28.9	22.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.324	0.381	0.442	0.501	0.499	0.517	0.508	0.475	0.448	0.395	0.354	0.326	(40)	
		taud	2.133	1.988	1.843	1.741	1.797	1.823	1.893	1.958	1.965	2.016	2.080	2.125	(41)	
	Ebn at Noon		257	256	252	246	251	246	247	251	248	247	241	244	(42)	
		Edh at Noon	34	46	59	70	68	66	61	56	52	44	36	32	(43)	
(44) All-Sky Solar Radiation	RadAvg		847	1175	1621	1989	2252	2266	2178	2040	1753	1390	962	753	(44)	
	RadStd		60	49	70	101	100	136	86	102	84	44	42	45	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.68	N/A	N/A	N/A	-0.79	N/A	N/A	-151	+198	+126
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-191	+151	+56

Nomenclature: See separate page